Week 10 Agentic Coding and Revision

FINS5545: Financial Market Data Literacy

What have we learned?

Agentic coding + turning raw data into insights ...

- Take raw data, both **structured** numbers like prices and **unstructured** text like news headlines.
- Clean the data – then use it for modelling – e.g., optimisation, sentiment, etc.
- Build it with **agentic coding tools**, where you direct an AI agent as opposed to manually writing code yourself.

Turn raw data into something useful. Build a narrative / create a useful application.

The course project tests these skills ...

For the assessment, this idea became a **portfolio management app**.

- The raw data was daily prices for 50 stocks and 10 cryptos, plus 149,683 news headlines.
- The “useful product” was a set of investable funds, each with a fact sheet, that a user can compare and allocate money across.
- Anyone can open it in a web browser, with no finance background needed.
- I encourage you to use your new skillset from this course and apply it to your own ideas / other courses at UNSW.

The project turned raw price data and news headlines into a product that offers investable funds to anyone through an app.

Today's (Final) Lecture

We use the project and revise the course (structured and unstructured data), and we focus on **Part B**, which is due on Friday of Week 11 – ignore any other date, this is the correct hand-in date.

- Structured part of the course, the out-of-sample optimal portfolios from Week 5, across stocks, crypto, and the two combined.
- Unstructured part, the news sentiment index and the fear and greed index from Weeks 8 and 9.
- We fold sentiment into the funds and ask whether it helps performance.
- We turn the results into a deployed app, **Overfit Capital** (my silly name for my app)

Every result today comes from the project reference code, so it is exactly what your own Part B can produce – you have most of the code to achieve similar results from this revision lecture.

How Part B Is Marked

Part B is worth 50 percent of the course. Within Part B, the marks split across six bands, taken from the project rubric.

Band	Share of Part B
Funds — optimal portfolios and OOS backtest (Station 3)	15%
Sentiment index and fusion extension (Station 3)	10%
Innovation and data-driven results	**30%**
Streamlit app and implementation (Station 4)	15%
Economic interpretation, reflection and writing	10%
AI workflow and transparency	**20%**
Total	100%

Innovation is the largest band at 30 percent and the AI workflow another 20 percent, so original work and a documented AI process is 50% of the final mark.

The Data Factory Floor

The project runs the four-stage **Data Factory Floor** from start to finish.

- **Station 1, Data Lake.** Load and clean the stock prices, the crypto prices, and the news headlines. (Part A)
- **Station 2, Feature Engineering.** Turn prices into returns and assemble the headlines into a daily text panel. (Part A)
- **Station 3, Model Design.** Build the out-of-sample funds, the sentiment index, and the fusion of the two. (Part B)
- **Station 4, Implementation.** Deploy the funds and the analytics as a Streamlit app. (Part B)

Part B is Stations 3 and 4, and today covers both, on the same 50 stocks, 10 cryptos, and 149,683 headlines your data ZIP contains.

The Product You Can Build

The app you build is an investment product – a user opens it, compares your funds, reads a fact sheet, and allocates money across them. The business (yours) earns a management fee (typically a small percentage).

- A **fund** is an asset universe combined with an optimisation method, for example *Combined Risk Parity*.
- A **fact sheet** reports growth of one dollar, return, volatility, the Sharpe ratio, the drawdown, and the current holdings.
- Every number a user observes is measured **out of sample**, so the weights that produced it were set on earlier data only.
- Your app gives the user an opportunity to invest in your funds – you make money from the management fee.
- Why are we doing this? Very few equity-crypto combined funds exist – we fill that gap in the market.

Overfit Capital is our worked example of that product. Your own app is the same idea under your own name.

The Pieces of Part B

A complete Part B brings together your final report hand-in, and a deployed app. The lists below are a guide to what fits together.

Exhibits worth building

1. A performance-metrics table across funds
2. Growth of one dollar
3. A drawdown figure
4. Portfolio weights over time
5. A Sharpe or risk-return barplot
6. A sector sentiment index over time

Result files the app (could) read in

- `fund_returns.csv`
- `fund_weights.csv`
- `sector_sentiment_index.csv`
- `performance_metrics.csv`

Part 1
Structured Data
Optimal Portfolios Out of Sample

Stage 3, Building Optimal Portfolios

A portfolio is a set of weights across assets. An **optimal portfolio** sets those weights to trade off return against risk, following Markowitz (1952). In this example, we build four funds from the same set of returns.

- **Equal-weight** puts one over N weight in every asset and uses no return or covariance data – easiest model
- **Minimum-variance** sets the weights that give the lowest portfolio volatility (in-sample).
- **Maximum-Sharpe**, or tangency portfolios, sets the weights with the best return per unit of risk (Sharpe1966) — again, in-sample.
- **Risk parity** gives every asset the same share of total risk (Maillard et al.2010), where risk is defined as volatility.

Each method is a 'simple' optimisation that turns average returns and a covariance matrix (how the assets move together) into weights. The hard part is that both inputs are estimated with error.

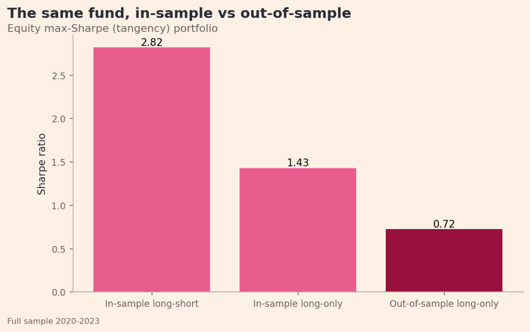
Optimisation functions

Each fund computes its weights w from the same two inputs, the average returns $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$. All are fully invested and long-only, so $1^\top \mathrm{w} = 1$ and $\mathrm{w} \geq 0$. You do not have to use long-only funds – you can use long-short if you want.

- **Equal-weight (1/N)** — no optimisation at all, just $w_i = 1/N$.
- **Minimum-variance** — the smallest portfolio variance, $\min_{\mathrm{w}} \; \mathrm{w}^\top \boldsymbol{\Sigma}\, \mathrm{w}$.
- **Maximum-Sharpe (tangency)** — the best return per unit of risk, $\max_{\mathrm{w}} \; \dfrac{\mathrm{w}^\top(\boldsymbol{\mu} - r_f 1)}{\sqrt{\mathrm{w}^\top \boldsymbol{\Sigma}\, \mathrm{w}}}$.
- **Risk parity** — every asset's risk contribution is equal, solve for w so that $\mathrm{RC}_i = \dfrac{w_i(\boldsymbol{\Sigma}\mathrm{w})_i}{\mathrm{w}^\top \boldsymbol{\Sigma}\, \mathrm{w}} = \dfrac{1}{N}$.

Testing a Fund Out of Sample

We care about out-of-sample returns, i.e., those actually earnable by your investors ... The equity tangency fund shows how large the divergence between in-sample (IS) and out-of-sample (OS) results is,



In-sample the tangency fund delivers a Sharpe of 2.82. The same fund, traded out of sample, delivers 0.72. The in-sample number cannot be traded – it contains 'look-ahead bias'.

The Walk-Forward Backtest (out-of-sample backtest)

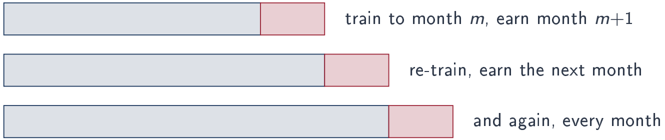
We evaluate every fund with a **walk-forward** backtest, which uses only data from before each decision – i.e., mimics how a fund would have traded in ‘real-time’, e.g., with monthly data.

- Estimate the inputs on the first year of returns, then set/estimate the weights.
- Earn the next month return at those weights, then re-estimate and rebalance.
- Repeat to the end of the sample.

Here the first live date is January 2021, the window expands each month, and there are 36 monthly rebalances through the end of 2023. We assume a risk-free rate of zero and annualise stocks and combined on 252 days and crypto on 365.

Re-Training on an Expanding Window

Each month the training window grows to include all data up to that point, an **expanding window**. We set the weights on that window, earn the next month, then roll forward and repeat.



the blue training window grows and the red earned month slides forward

Stitching these earned months together gives a continuous out-of-sample track record. On this data that is 36 monthly re-trainings.

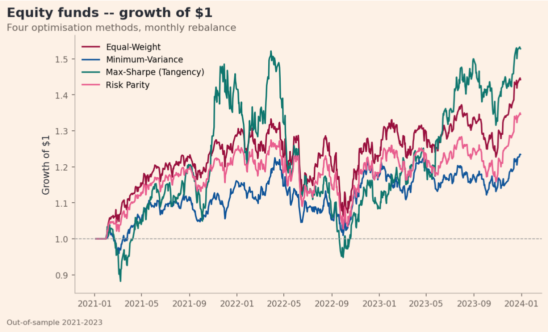
Equity, Crypto and Combined Data

We run the same four methods on three investment universes.

- **Equity** is the 50 stocks across 10 sectors.
- **Crypto** is the 10 cryptocurrencies.
- **Combined** joins the crypto returns onto the stock trading calendar, so a fund that trades on stock-market days could actually hold both.

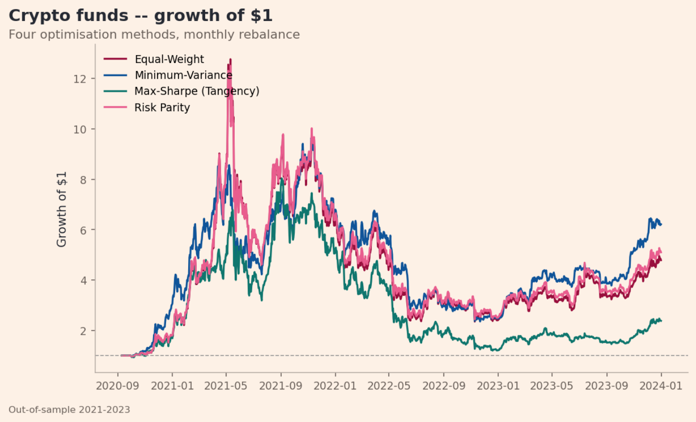
Growth of One Dollar, Equity

Each line is a fund, plotted as what one dollar grows to, so a line ending higher earned more and a dip along the way is a loss.

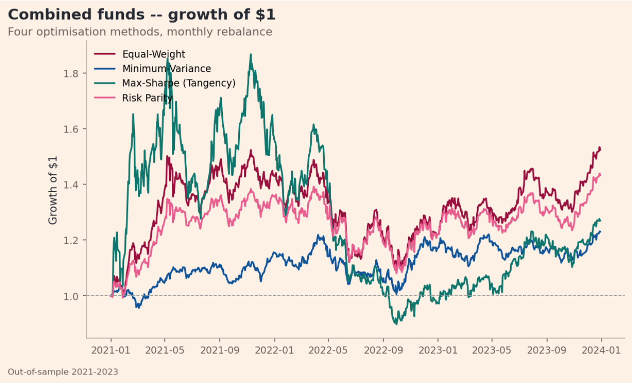


Equal-weight and risk parity fall less in the 2022 drawdown than the concentrated tangency fund.

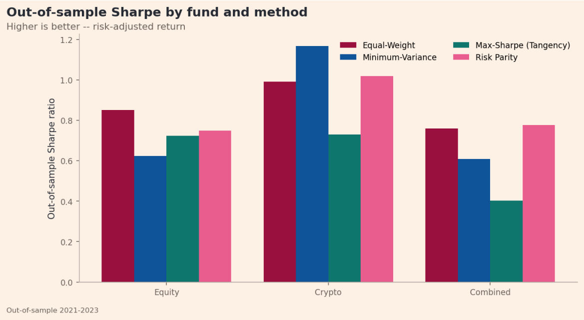
Growth of One Dollar, Crypto



Growth of One Dollar, Combined



The Funds Ranked by Sharpe

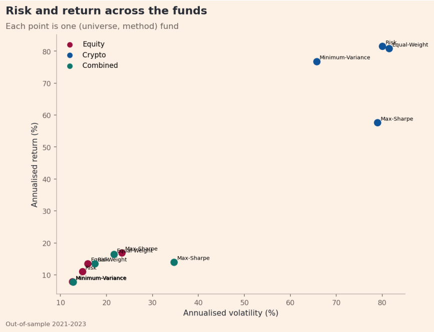


Over 2021 to 2023 crypto had the best risk-adjusted return, so the crypto funds rank highest and the combined funds rank below crypto alone.

Diversification is regime-dependent. Over this window crypto performed unusually well, and mixing it with stocks lowered its Sharpe.

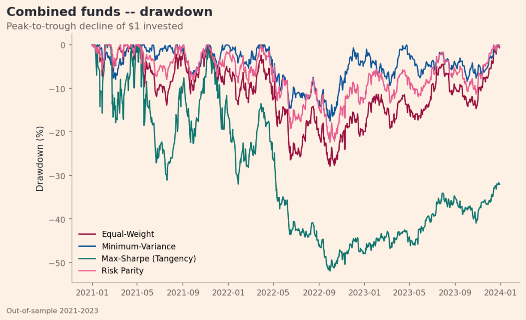
Combined Risk and Return

Each dot is a fund, placed by its volatility, or risk, on the horizontal axis and its annual return on the vertical axis, so dots further right are riskier and dots higher up returned more.



The crypto funds have high return and high volatility, and the stock funds cluster at low risk. A user could pick a point on this map by their appetite for risk.

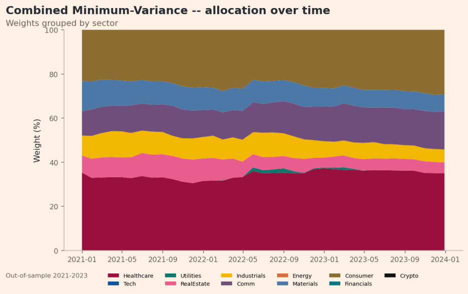
A **drawdown** is how far a fund has fallen below its highest earlier value. The maximum drawdown is the worst such fall over the whole period.



The maximum drawdown of the combined tangency fund reaches 52 percent. Minimum-variance holds its worst loss near 18 percent. A fact sheet must show the drawdown as well as the growth line.

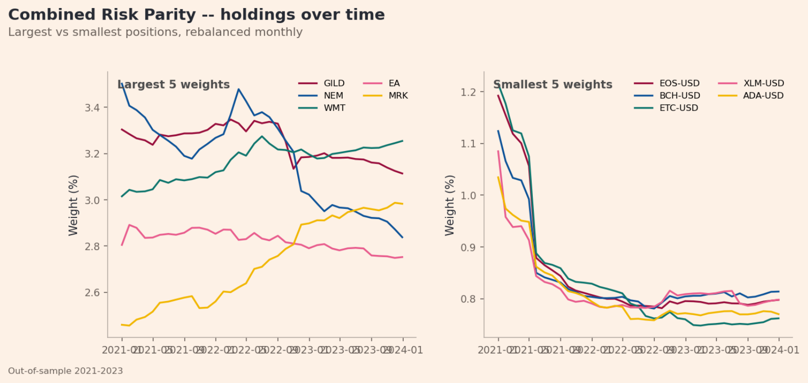
Portfolio Weights Over Time

Each band is a sector's share of the fund, with crypto shown as a single black band, and the bands stack to 100 percent on every date, so a widening band means the fund holds more of that sector.



Minimum-variance concentrates in defensive sectors, mainly healthcare (about 34 percent) and consumer (about 25 percent), and holds almost no crypto, because minimising variance avoids the most volatile assets. The weights move each month as the covariance matrix is re-estimated.

Largest and Smallest Holdings



Risk parity holds every asset, but its largest weights are stable low-volatility stocks and its smallest are the volatile cryptos. The method gives each asset an equal share of total risk, so the safest assets get the largest dollar weights.

Fund performance

Out-of-sample Sharpe ratio, which is annual return divided by volatility so higher is better, for all 12 funds. Return, volatility, and drawdown are shown in the app.

Method	Equity	Crypto	Combined
Equal-Weight	0.85	0.99	0.76
Minimum-Variance	0.62	**1.17**	0.61
Maximum-Sharpe (Tangency)	0.72	0.73	0.40
Risk Parity	0.75	1.02	**0.78**

Equal-weight is hard to improve on among the stock funds, which matches DeMiguel et al. (2009), who find that no “clever rule” reliably improves on one over N out of sample.

The best combined fund is risk parity at 0.78. A method that loses to equal-weight, explained clearly, is still good project work.

Ideas, Structured

The 12 funds are a potential baseline – going beyond them is where the innovation marks are earned.

- Add a tail-aware objective such as mean-CVaR, which targets the worst outcomes in the tail of the returns.
- Reduce estimation error with covariance shrinkage.
- Add volatility targeting, which scales exposure up and down with recent risk (Moreira and Muir2017).
- Blend equal-weight with minimum-variance.
- Add a turnover or transaction-cost model.
- Dozens of other potential ideas ...

Pick one or more extensions, motivate them, and show their out-of-sample effect. A careful extension is worth more than several shallow ones.

Part 2
Unstructured Data
Sentiment From the News Headlines

News headlines

Sentiment can move prices – the unstructured part of the course focused on textual data. Tetlock (2007) links pessimistic news to lower prices, and Baker and Wurgler (2007) show investor sentiment is an important risk factor.

- Your data has 149,683 dated headlines for the 50 stocks over 2020 to 2023.
- Station 2 assembled them into a daily text panel, one row per stock per day.
- Station 3 uses VADER or your own VADER model to produce sentiment scores.

The news is headlines only, which is a noisy proxy for the full article. Say so in your report, and treat a compound score of zero as potentially missing information.

VADER, the Project Baseline

VADER is a rule-based sentiment model from Hutto and Gilbert (2014). It scores a sentence and returns a **compound score** between minus one and plus one, where minus one is the most negative, zero is neutral, and plus one is the most positive.

- It scores words from a dictionary/lexicon and adjusts for capitals, punctuation, boosters, and negation (VADER rules).
- You do not clean the text first, because lower-casing and stripping punctuation remove the data VADER uses.
- **finVADER** adds finance word lists, so it captures market language that plain VADER does not.
- I strongly encourage you to use your AI systems and build your own lexicon or your own VADER model.

From Headlines to a Sector Index

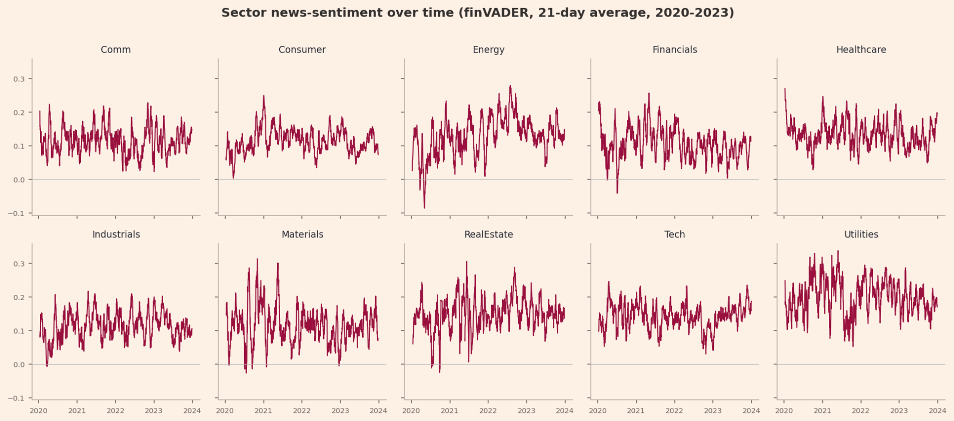
We build the sector sentiment index in three steps.

- Score every headline, then average the scores for each stock on each day.
- Average the stock scores within each sector, giving every stock equal weight. Can you think of other weightings that are not 1/N?
- Lag the signal by at least one trading day, so a Monday headline is first usable on Tuesday.

The lag is what keeps the index look-ahead safe. A day-t decision uses only sentiment from day t minus one or earlier.

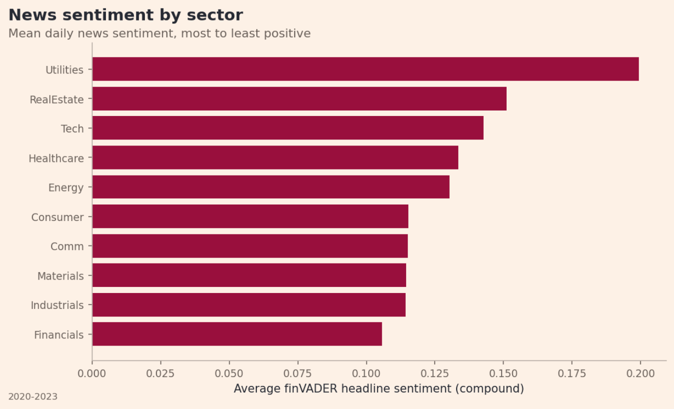
Equal-weight the stocks, decide how to treat days with no news, and lag the signal. Those three choices are important, so state them in your report.

Sector Sentiment Over Time



Every sector's sentiment is above zero most of the time, and all of them fall together at the COVID crash in early 2020. News sentiment has a positive baseline.

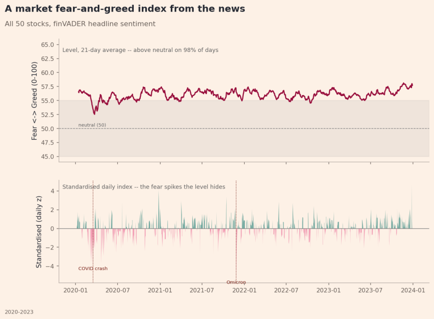
Sectors Ranked by Sentiment



Utilities and real estate news is the most positive over 2020 to 2023, financials the least.

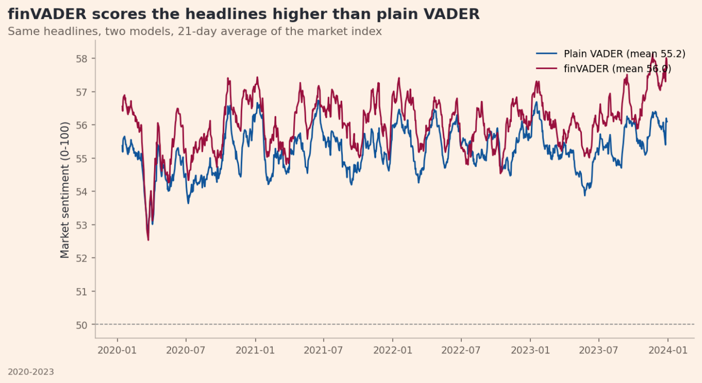
A Fear and Greed Index

Average the sentiment across all 50 stocks, rescale it onto 0 to 100, and standardise it (subtract the mean and divide by the standard deviation).



The raw level is above neutral on 98 percent of days, so it always reads as greedy. Standardising is what reveals the fear spikes, the deepest being the Omicron variant announcement (COVID) and the COVID crash itself.

Comparing VADER and finVADER



The same headlines receive different scores under two models. The finance lexicon scores market news about two points higher on the 0 to 100 index and leaves fewer headlines as neutral. Try to develop your OWN custom VADER model.

Some Ideas – Unstructured Data

The sentiment model is the most open part of the project.

- Extend the VADER lexicon with finance terms, then have your AI agent rate them and keep the ones raters (AI agents) agree on.
- Swap in TextBlob, which returns polarity and subjectivity, or blend two models into an ensemble.
- Build the index by market, by sector, or by individual stock, and try different smoothing windows.
- Use an expanding-window standardisation for a live index, and the full-sample version for history.

A finance lexicon extension, built and tested, is a strong and expected innovation route.
Keep a record of how you rated the terms with your AI agents.

Part 3
Fusion
Sentiment Tilts (Idea) for the Funds

Including Sentiment Into a Fund – Example with the Minimum-Variance Fund

A **sentiment tilt** nudges the minimum-variance equity fund toward stocks with better-than-usual news and away from stocks with worse news. We keep the fund weights we already have and move them only a little, based on the news sentiment.

- We start from the base weights w_i^{base}, the minimum-variance equity weights from earlier.
- We use each stock's daily news sentiment score $s_{i,t}$, already built in the sentiment step.
- We shift weight toward the good-news stocks and away from the bad-news stocks, controlled by a tilt strength λ.

A positive λ is **momentum**, holding more of the good-news stocks. A negative λ is **contrarian**, holding more of the bad-news stocks. At $\lambda = 0$ nothing moves, and the fund is exactly the base.

The baseline stays the minimum-variance equity fund, and we tilt non-parametrically toward high-sentiment or low-sentiment stocks.

The Sentiment Tilt – Step by Step

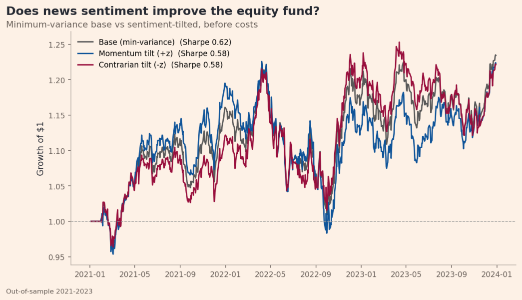
Three steps turn the base weights w_i^{base} and the daily sentiment scores $s_{i,t}$ into the tilted weights for day t.

1. **Standardise** each stock's sentiment into a rolling **z-score** over the past N days, so a value above zero means better-than-usual news,
$$z_{i,t} = \frac{s_{i,t} - \bar{s}_{i,t}}{\sigma^s_{i,t}}.$$
2. **Tilt** each base weight up or down in proportion to that z-score,
$$\tilde{w}_{i,t} = w_i^{\text{base}} \left(1 + \lambda\, z_{i,t}\right).$$
3. **Clip and renormalise** so the fund stays long-only and fully invested,
$$w_{i,t} = \frac{\max(\tilde{w}_{i,t},\, 0)}{\sum_j \max(\tilde{w}_{j,t},\, 0)}.$$

Here $\bar{s}_{i,t}$ and $\sigma^s_{i,t}$ are the rolling mean and standard deviation of the score, and λ is the tilt strength. The z-score uses news only up to the day before, so there is no look-ahead bias.

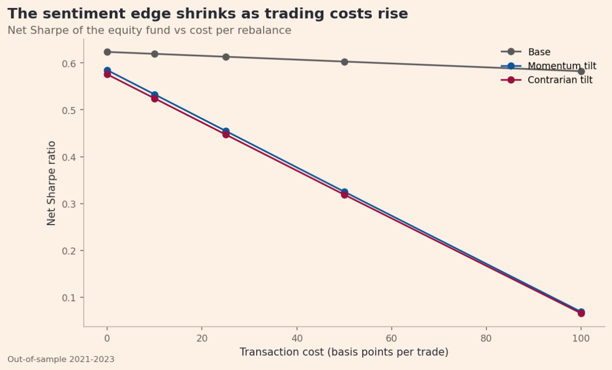
A Naive Tilt Does Not Improve Much on the Baseline Model

The base is the minimum-variance equity fund. The **momentum** tilt uses $\lambda = +1$ and the **contrarian** tilt uses $\lambda = -1$, both set by hand with no tuning, out of sample and before costs.



Base Sharpe 0.62, momentum 0.58, contrarian 0.58. A hand-set tilt in either direction scores below the base. A clear negative result, explained, still earns you marks.

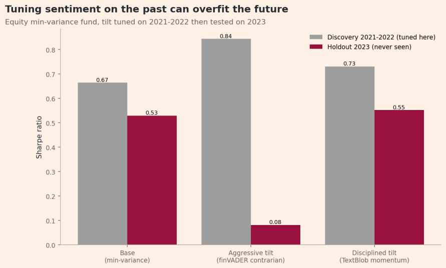
Trading Costs



A tilt trades more than the base, so trading costs cut into it faster. By 50 basis points per trade any gain the tilt might have had is gone.

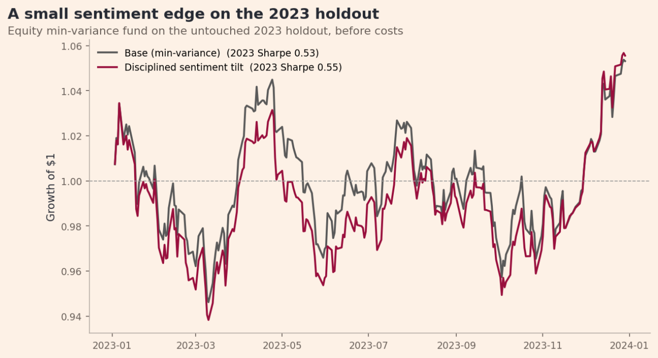
Tuning on the Past Can Overfit the Future

To **tune**, we try many sentiment choices — the model, the direction, and the strength λ — and keep whichever does best on 2021–2022 data, the **discovery** window (grey bars). We then test that choice out-of-sample using 2023 data (red bars).



The aggressive tilt (finVADER contrarian, $\lambda = -2$) topped tuning at 0.84 then fell to 0.08 on 2023 — a strong tilt can fit the past by luck and then fail on new data.

A Small Sentiment Edge Out of Sample



A milder TextBlob momentum tilt, chosen on the tuning years, lifts the 2023 holdout Sharpe from 0.53 to 0.55 before costs. The edge is small and fragile — it holds only before costs and only with a mild tilt.

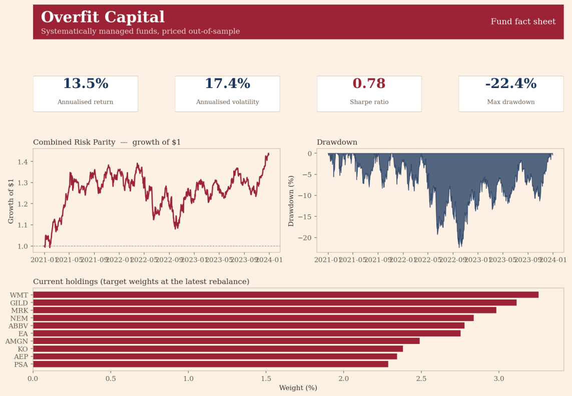
Part 4
Stage 4
Building and Deploying the App

Stage 4, From Results to a "Product"

The backtests and the sentiment index / tilts provide the results for the report write-up. Station 4 serves them to a user as a Streamlit app.

- The app loads the precomputed files and shows the fact sheets. It runs no backtest and loads no sentiment model.
- That keeps it light enough for the free tier, which has about one core.
- The folder is its own GitHub repository, and the entry point is `streamlit_app.py` at the folder root.

Precompute the slow work, save it, and have the app only load it. An app that recomputes on every click can time out on the free tier.



This is a fund's fact sheet inside the app. The same four numbers, growth line, drawdown, and holdings a fund provider would publish.

What a User Can Do

A good app lets a user do four things without help.

- **Compare funds.** A metrics table and a Sharpe chart across all 12 funds.
- **Read a fact sheet.** Growth, drawdown, and the current holdings for a chosen fund.
- **Read the sentiment analytics.** The fear and greed gauge, the sector index, and the sentiment tilt results.
- **Build an allocation.** Sliders to blend funds, a fee, and the net growth of the blend.

The Agentic Coding Example

The analysis is built. The deploy follows a set protocol, so you hand it to your AI agent with a prompt like this one.

```
Part B analysis is done - funds, sentiment index, fusion, and
all results/ CSVs and figures are built. Now follow the
fins-agent Part B protocol to build Station 4. First read
AGENTS.md and docs/STUDENT_DEPLOY.md in this folder.
- Build streamlit_app.py at the folder root that LOADS the
  precomputed results/ only (no backtest, no nltk at runtime).
- Load raw data via src/data_access.py, never commit it.
- Run scripts/check_handin.py and fix every failure.
- git init, commit the code and results/, push a NEW public repo.
The browser deploy on Streamlit is my step - it needs my login.
```

The agent follows the protocol files in your folder. Read what it writes, test it, and fix its mistakes — that reading and fixing is the graded skill.

Who Does What

Deploying splits cleanly between your agent and you.

- **Your agent** builds the app, runs `check_handin.py`, commits your code and the results files, and pushes to a new public repository.
- **You** open share.streamlit.io, sign in, and connect the repository. That step needs your own login, so the agent cannot do it.

The agent prepares everything up to the browser step, and you click deploy — neither half of the hand-off works without the other.

Deploy, Step by Step

1. Commit your code and the precomputed `results/` files. The app loads them, so they must be in the repository.
2. Run `streamlit run streamlit_app.py` and confirm it works on your machine.
3. Run `check_handin.py` and fix every failure it reports.
4. Push the folder to a new, public GitHub repository.
5. On share.streamlit.io, connect the repository with entry point `streamlit_app.py`.
6. At hand-in, confirm the live link opens in a logged-out browser.

The most common lost mark is a live link a marker cannot open. Re-test the public link in a logged-out browser before the deadline.

Your Public Repository

Your Part B folder is a **public** GitHub repository, so a marker can open the live app, read the code, and run it from a clean checkout.

- Commit your code and the precomputed `results/` the app reads.
- Load all raw data through the data-access helper, and never commit the raw price or news files.

Public repository, live link, and the zipped folder are the three things you submit for Station 4.

Document How You Used AI

Your AI workflow is graded at 20 percent of Part B, so keep records ...

- Replace the starter stubs with the agent files you actually used, and submit them.
- Keep a prompt log that shows the prompt, the AI output, and your own fixes / guidance.
- Have you made any skills/rules or agent files?
- Write the analysis and the economic interpretation in your own words.

A good entry reads like a short story about a problem. You asked for a backtest, the agent added a look-ahead bug, you caught it, and you record how.

AI makes the code easy, so the marks move to judgement. Show where the AI was wrong and what you did instead.

Finishing Part B

Ideas Across the Project

Innovation is rewarded across the whole project. Any of these, built and evidenced, reaches the top band.

Funds and data

- A new objective or a new factor
- Shrinkage or volatility targeting
- A transaction-cost model

Sentiment and app

- A finance lexicon or an ensemble
- A disciplined, tested sentiment tilt
- An original design or a useful feature

A careful extension that does not improve on the baseline, explained well, still earns you the band. The credit is for original, evidenced work, whatever the final number.

Finish the Project, a Checklist

- Build the combined fund plus the equity-only and crypto-only funds across several methods.
- Produce all seven exhibits and save the four named files.
- Build the sector sentiment index, lag it, and test a fusion tilt before and after.
- Deploy the app from a public repository with a live link, reading precomputed results.
- Write the report in your own words, interpret every exhibit, and give three concrete recommendations.
- Submit your agent files and prompt logs.

Part B is due Friday of Week 11. Everything on this list is something you saw built today.

Summary

- Optimal portfolios are only as good as their out-of-sample test, and equal-weight is a strong benchmark.
- A news sentiment index turns headlines into a fear and greed gauge across the market and its sectors.
- Sentiment can tilt a fund, but only a disciplined tilt leaves a small edge out of sample and after costs.
- The app serves the funds to a user, and one agent prompt builds and prepares the deploy.

Overfit Capital brings the entire course together as a working product. Your Part B is the same structure, built on your own choices and name.

Final Thoughts

"It ain't what you don't know that gets you into trouble. It's what you know for sure that just ain't so."

— adapted from Josh Billings (often misattributed to Mark Twain)

"If you torture the data long enough, it will confess."

— Ronald H. Coase

References I

Malcolm Baker and Jeffrey Wurgler. Investor sentiment in the stock market. *Journal of Economic Perspectives*, 21 (2):129–152, 2007. doi: 10.1257/jep.21.2.129.

Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, 22(5):1915–1953, 2009. doi: 10.1093/rfs/hhm075.

Clayton J. Hutto and Eric Gilbert. VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1):216–225, 2014. doi: 10.1609/icwsm.v8i1.14550.

Sébastien Maillard, Thierry Roncalli, and Jérôme Teiletche. The properties of equally weighted risk contribution portfolios. *The Journal of Portfolio Management*, 36(4):60–70, 2010. doi: 10.3905/jpm.2010.36.4.060.

Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.1111/j.1540-6261.1952.tb01525.x.

Alan Moreira and Tyler Muir. Volatility-managed portfolios. *The Journal of Finance*, 72(4):1611–1644, 2017.

William F. Sharpe. Mutual fund performance. *The Journal of Business*, 39(1):119–138, 1966. Part 2: Supplement on Security Prices.

Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3):1139–1168, 2007. doi: 10.1111/j.1540-6261.2007.01232.x.